Provisional DAC test equipment

Separate attachment for the buffered six bit DAC proposal | 12 November 2026

Team name: [TEAM NAME] ICARUS team number: [TEAM NUMBER]

Planning basis: all equipment and lab access below are assumed available for the initial proposal. Instrument models, calibration status, bookings and access owners are to be confirmed before measurement. These are required capabilities, not a claim that particular instruments have already been checked.

Equipment or access	Required capability	Use
Low noise DC supplies	Independent current-limited 3.3 V and 1.8 V rails, adjustable across proposed limits; current readback or external metering.	Analogue/digital power and supply sweeps.
Two channel SMU or precision references with source and sink support	0.3 V and 3.0 V outputs; ≤1 mV set/measurement uncertainty after calibration; at least ±0.2 mA compliance.	Drive VREFH and VREFL, including sinking ladder current; measure reference loading.
Precision bias source	1 µA nominal current into IBIAS, adjustable 0.8–1.2 µA with suitable compliance; or approved harness bias.	Standalone bias tests and reference cross-check.
Calibrated precision DMM	6½ digits or equivalent; combined voltage uncertainty ≤1 mV over 0.3–3.0 V. Low-current measurement or SMU support.	Code sweep, static error, linearity and current measurements.
Digital oscilloscope	At least two simultaneous channels, ≥100 MHz bandwidth, ≥500 MS/s; waveform export and analysis of long settling records.	VOUT and EN_MON timing; live-update glitches and noise characterisation.
Low capacitance active probe	Input capacitance preferably ≤2 pF, high DC input resistance, appropriate input range.	Observe VOUT while maintaining the total 20 pF load budget.
Harness controller or FPGA interface	Set six data controls and EN at 1.8 V, observe EN_MON test route, provide deterministic updates and automated readback.	Functional sweeps, reset protocol and transient stimulus.
Temperature chamber or calibrated thermal stage	−40 to 110 °C capability, package temperature monitoring and safe feedthroughs.	Temperature characterisation of transfer, settling and current.
LCR meter and precision passives	Resolve fixture capacitance; calibrated 100 kΩ and 1 MΩ loads, C0G capacitors and stress loads.	Load fixtures, probe/board capacitance budget and loading tests.
Workstation and IC laboratory access	Sky130 PDK; Xschem/ngspice, Magic/KLayout, Netgen and extraction tools; board assembly and controlled handling facilities.	Open-source verification, layout/sign-off, scripts and test-board bring-up.

Fixture plan: use separate power/current sensing, short reference wiring and a low capacitance probe. Characterise output loading and EN_MON delay before timing tests. The lower reference must sink current. Record access and calibration details before measurement. Designer CVs are separate attachments; populate team placeholders before submission.